AMME2000/BMET2960/BMET9960 - Assignment 1, 2026

Due: 11:59 pm Thursday 2nd April (Week 6) 2026

Note: The official deadline for Assignment 1 is Thursday April 2nd @ 11:59 pm. However, **a Simple Extension will be automatically applied for all students, so you may submit Assignment 1 up until Tuesday April 7th @ 11:59 pm without a late penalty**. The official due date will remain unchanged on Canvas - so you cannot apply for a simple extension on top of the one already given. **Please do not submit a Simple Extension request since it is automatically applied**.

Assignment 1 focuses on the analytical and numerical solution to the Heat/Diffusion Equation.

Your report must be your own work. Please see the **FAQ on Academic Integrity** on our Canvas site for information on what is permitted and not permitted for Assignments. You must also comply with the **AMME2000/BMET2960/BMET9960 Generative AI Policy**. You must also submit a completed **Generative AI Use Statement** with your Assignment.

Any students who use paid/subscription services to post and/or view assignment questions, or who contract a third party to complete all/part of their Assignment will receive 0 and will be reported for academic misconduct. For more information, see the first lecture where academic integrity, including penalties, was discussed at length.

IMPORTANT INFORMATION FOR YOUR REPORT

- It MUST be **PDF** format
- It MUST be **typed** (including all math working). Handwritten assignments receive 0.
- It MUST insert all **plots/figures as images** NOT screenshots. 10% penalty for screenshots.
- It MUST be **≤10 pages**. Page limit excludes the Appendix. **DO NOT copy the assignment questions into your report, only include your answers**.
- It MUST include your **MATLAB code in an Appendix** to the assignment report.
- It MUST be **submitted by the deadline** to avoid late penalties (5%/day).

Structure your report according to the Section and Question numbers (e.g. 1.1, 2.2) and include all relevant working and answers in the appropriate section. Any figure / table in your report should have a label and be referred to and discussed.

We have prepared an **exemplary assignment** to guide you in writing your own.

INTRODUCTION: GLACIAL MELTING

Glaciers are large natural structures of ice and rock which typically have complex temperature profiles governed by environmental exposure, geothermal effects and mechanical pressure. We will consider only the environmental effects and model the temperature profile in a simplified 1D glacial section using the heat/diffusion equation:

$$\frac{\partial T}{\partial t} = D\frac{\partial^2 T}{\partial x^2} \tag{1}$$

where $T = T(x,t)$ is the temperature along the glacial section at location x and time t, D is the thermal diffusivity constant, and the spatial domain is $0 \leq x \leq L$ metres (**Figure 1**).

Figure 1. The glacial section of length L.

One end of the glacial section is exposed to the environment which we can assume has a constant average temperature of $2\ ^\circ\mathrm{C}$. The other (basal) end of the glacial section is permanently frozen at $-40\ ^\circ\mathrm{C}$. Assume that the entire glacial section is initially at the basal temperature.

QUESTION 1: ANALYTICAL SOLUTION [50 mk]

Relevant parts of the course to review for this question:
Week 2 Lecture 2, Week 3 Lecture 1, Week 4 Tutorial, Weekly Practice Problems

1.1 If the thermal conductivity of the glacial ice is 2.6 watts per metre-Kelvin, the specific heat capacity is 1800 joules per kilogram per Kelvin, and the density is 903 kilograms per cubic metre, show that:

(i) The magnitude of D is 1.60×10^{-6} **[1 mk]**

(ii) The units of D are $\mathrm{m}^2.\mathrm{s}^{-1}$ [**1 mk**]

1.2 Boundary conditions (BCs) and initial condition (IC):
(i) State the BCs of the problem mathematically. **[4 mk]**
(ii) State the IC of the problem mathematically. **[2 mk]**

1.3 Briefly explain why we *cannot* assume the solution to (1) has the form:

$$T(x,t) = F(x)G(t) \tag{2}$$

where $F(x)$ and $G(t)$ describe the spatial and temporal variation, respectively. **[2 mk]**

1.4 Similarly to the heat pipe example given in the lectures, to obtain the solution $T(x,t)$ for the glacier problem we must start by assuming it has the form:

$$T(x,t) = T_S(x) + T_H(x,t) \tag{3}$$

(i) Briefly explain what $T_S(x)$ refers to and what form you expect $T_S(x)$ to have given the problem description. **[2 mk]**

(ii) Derive $T_S(x)$, showing your working. Provide a labelled sketch of $T_S(x)$. **[5 mk]**

1.5 With reference to $T_H(x,t)$ in (3):

(i) Briefly explain what $T_H(x,t)$ refers to. **[2 mk]**

(ii) Determine the IC for $T_H(x,t)$, showing working. Provide a labelled sketch of this 'modified' IC. **[4 mk]**

1.6 Using separation of variables for $T_H(x,t)$ we obtain the following ODEs describing the spatial variation $F(x)$ and temporal variation $G(t)$:

$$F_{xx} + p^2 F = 0 \tag{4a}$$

$$G_t + Dp^2 G = 0 \tag{4b}$$

Solve these ODEs to show that the solution for T_H is given by:

$$T_{H,n}(x,t) = \sum_{n=1}^{\infty} B_n^* \sin\left(\frac{n\pi x}{L}\right) e^{-(\lambda_n)^2 t} \tag{5}$$

In your working ensure that all relevant terms are defined. **[5 mk]**

1.7 Use your solutions to 1.5ii and 1.6 to solve for the Fourier coefficients B_n^* in (5). **[2 mk]**

Hint: Here you may use the fact that the Fourier sine coefficients for a triangular function $h(x)$ of the form shown in Figure 2 are given by:

$$H_n = \frac{2A}{n\pi}, n = 1, 2, 3, \ldots \tag{6}$$

i.e. you do NOT need to use the Euler formulae on the Data Sheet and integrate by parts.

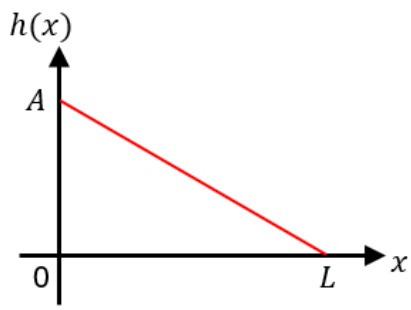

Figure 2. Triangular function.

1.8 Hence show that the complete solution for the evolving temperature in the glacial section is given by:

$$T(x,t) = \sum_{n=1}^{\infty} \left(-\frac{84}{n\pi} \sin\left(\frac{n\pi x}{L}\right) e^{-(\lambda_n)^2 t} \right) - \frac{42x}{L} + 2 \tag{7}$$

[2 mk]

1.9 Use MATLAB to implement the analytical solution in (7) for a glacier of length $L = 256$ m.

Hint: Write a function that computes your analytical solution with n_x (number of grid points in x), n_t (number of grid points in time) and n_F (number of Fourier terms) as input variables, and with the analytical solution vector as the output variable. This will enable you to change relevant input variables easily and generate the solution vector. We strongly recommend including a plot command inside the time loop of your function so that you can animate the solution at each time step and visualise the behaviour over time. By adding a 'plot flag' input parameter to your function, you can easily turn the animated plotting on/off: setting the flag to 0 = no plotting; setting the flag to 1 = plot analytical solution at each time step. Use your function to generate the solution vector for required input parameters and then use the 'hold on' command to overlay these solutions on the same plot. You may also find the MATLAB *axis* function useful to limit your plot to a region of interest in x and/or T for clearer visualisation.

Answer the following:

(i) **Figure 3** shows the analytical solution at $t = 12$ h using $n_F = 10$, $n_F = 100$, $n_F = 250$, $n_F = 500$ and $n_F = 1000$. (Note that the vertical and horizontal plot range has been adjusted to focus on the exposed end of the glacial section.)

Briefly discuss the plot in Figure 3 and whether or not it appears to be correct. **[2 mk]**

Propose a practical value for n_F if we are mainly interested in the temperature profile of the glacier beyond 10 days. Justify your answer. **[2 mk]**

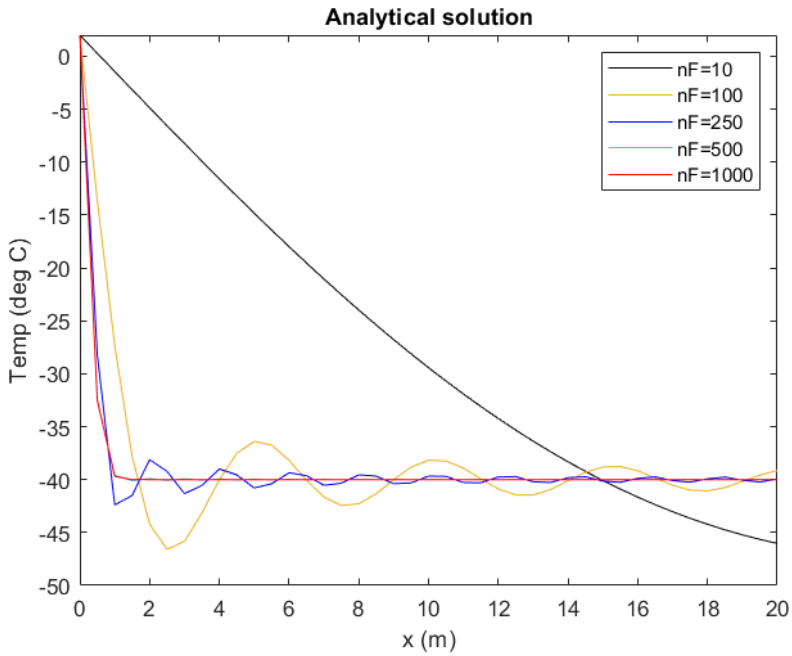

Figure 3. Analytical solution of the glacier temperature profile after 12 hours for varying n_F.

(ii) Use your MATLAB code to generate the analytical solution of the temperature profile along the entire glacial section at $t = 10$ d, $t = 100$ d, $t = 1000$ d, and $t = 10{,}000$ d and plot these profiles together on the same figure with a legend distinguishing them. For this simulation use $n_x = 513$ and your proposed value of n_F from Q1.9i. **[6 mk]**

In your Report, include a **hyperlink** here to the relevant MATLAB code in the Appendix

(iii) Briefly discuss the temperature behaviour in your plot from 1.9ii. **[2 mk]**

(iv) Determine the length of glacier that has melted at $t = 10$ d, $t = 100$ d, $t = 1000$ d, and $t = 10{,}000$ d. Your answer should be to the nearest 0.5 m in each case (i.e. the resolution of the spatial grid). In your report, include a brief explanation of your approach. **[4 mk]**

In your Report, include a **hyperlink** here to the relevant MATLAB code in the Appendix

(v) Now repeat Q1.9iv assuming the environmental temperature (i.e. the first BC) has risen by 1 °C and again compute the length of melted glacier at the four time points. **[2 mk]**

QUESTION 2: NUMERICAL SOLUTION [30 mk]

Relevant parts of the course for this question:
Week 4 Lecture 1, Week 5 Lecture 1, Week 5 Lecture 2, Week 5 tutorial

Using a forward-in-time (FT) approximation for the temporal derivative and a central-in-space (CS) approximation for the spatial derivative, the governing equation (1) can be discretised to give an explicit numerical expression for the temperature:

$$T_i^{n+1} = T_i^n + \frac{D\Delta t}{(\Delta x)^2}(T_{i+1}^n - 2T_i^n + T_{i-1}^n) \tag{8}$$

where D is the diffusivity.

2.1 Briefly define each of the following terms in (8) for the glacier problem: **[5 mk]**
(i) i
(ii) n
(iii) T_i^n
(iv) Δx
(v) Δt

2.2 With reference to (8) and the space-time diagram in **Figure 4**, briefly explain which node(s) we could compute the temperature for given the temperature at the nodes circled in red. Assume $L = 256$ m, 513 spatial grid points, and $\Delta t = 43200$ s. Also assume 1-indexing as used in MATLAB implementations. Your answer must include the specific location and time of the node(s). (Note that you are NOT required to compute an actual temperature.) **[2 mk]**

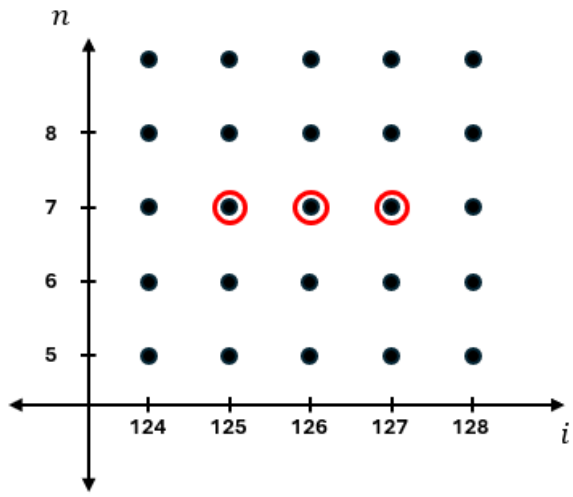

Figure 4. Space-time diagram example.

2.3 Now implement the numerical solution in MATLAB.

Hint: Extend your function from Q1.9 so that it also computes the numerical solution. Include both the analytical and numerical solution vectors as function outputs. You can extend the 'plot flag' input parameter to allow for 4 options: 0 = no plotting; 1 = plot analytical solution only; 2 = plot numerical solution only; 3 = plot both the analytical and numerical solution – enabling you to check and visualise the behaviour of the different solutions over time.

Use your code to plot the numerical solution at $t = 10$ d, $t = 100$ d, $t = 1000$ d, and $t = 10000$ d. Use 513 grid points and $\Delta t = 43200$ s. Present a plot of the 4 overlaid curves in your report, including a legend. **[4 mk]**

In your Report, include a **hyperlink** here to the relevant MATLAB code in the Appendix

2.4 This question focuses on the stability of the numerical solution.

(i) Calculate the maximum stable time step for the numerical scheme if $n_x = 513$. Show your working. **[2 mk]**

(ii) By choosing a time step slightly larger than the maximum stable value, use your simulation to demonstrate the transition from stability to instability by plotting the analytical solution overlaid with the (unstable) numerical solution. Choose your simulation time so that the error due to instability is clearly visible but not excessive. Make sure to state in your report what parameters you used. You may want to use the *axis* command to limit the x-axis plot range for better visualisation. **[3 mk]**

2.5 **Table 1** is used to record error norm data (L^1, L^2 and L^∞) and to calculate the resulting convergence rate ($\mathcal{O}(L^1)$, $\mathcal{O}(L^2)$, $\mathcal{O}(L^\infty)$) for the numerical scheme. (This table follows the example shown in the lectures.)

Table 1. Error norms and order of accuracy

n_x	L^1	L^2	L^∞	$\mathcal{O}(L^1)$	$\mathcal{O}(L^2)$	$\mathcal{O}(L^\infty)$
65						
129						
257						
513						
1025						

(i) Use your MATLAB simulation with $n_t = 1251$ and $\Delta t = 600$ s to compute the error norm and convergence rate data for **Table 1** (shaded boxes) and present the completed table in your report. Use 6 sig figs to represent your tabulated values. (See Appendix for relevant formulae.) **[4 mk]**

In your Report, include a **hyperlink** here to the relevant MATLAB code in the Appendix

(iii) Briefly comment on whether the convergence rate data behave as you expect, justifying your answer. **[2 mk]**

2.6 This question focuses on the choice of discretisation for the numerical solution.

(i) Plot your numerical solution of temperature along the glacier at $t = 10$ days using 5 different choices of spatial gridding: $n_x = 65, 129, 257, 513$. Use $\Delta t = 43200$ s for all simulations to ensure stability of the numerical solution. Overlay the 4 different numerical curves on a single plot and also overlay your analytical solution from Q1 (use $n_x = 513$ and 250 Fourier terms for the analytical solution). Your plot should therefore have 5 curves and include a clear legend to distinguish them. Restrict your plot to the 'heel' of the curve by using the command *axis([0 10 -40 -35])*.

Present your labelled plot in your report. **[6 mk]**

In your Report, include a **hyperlink** here to the relevant MATLAB code in the Appendix

(ii) Briefly discuss your plot from 2.6i and what it implies about the grid choice. **[2 mk]**

QUESTION 3: BONUS [10 mk]

Note: Question 3 is for bonus marks and is optional to attempt. Students can achieve 100% in the assignment by completing Questions 1 and 2; they can achieve >100% by completing Question 3.

3.1 The scenario described in Q1 and Q2 assumes that those parts of the glacier which melt remain connected to the glacial section, such that L remains constant. In reality, the parts of the glacier which melt would be removed as 'meltwater', meaning that L gets smaller over time. Adapt your code to account for this change in L over time and generate a plot of the glacial temperature profile at $t = 10$ d, $t = 100$ d, $t = 1000$ d, and $t = 10000$ d. Overlay your previous plots (which assume constant $L = 256$ m) for comparison. Use $\Delta t = 43200$ s and 513 grid locations for the glacier. **[6 mk]**

In your Report, include a **hyperlink** here to the relevant MATLAB code in the Appendix

Procedure and hints:

- Since the spatial domain is discretised (not continuous), each node (grid point) represents a 'slab' of ice. Slabs will be 0.5 m thick since $L = 256$ m and $n_x = 513$. **You should retain the same discretisation $\Delta x = 0.5$ as ice melts.**
- Assume that a slab melts away when the temperature of the node is greater than 0 °C.
- Once a slab melts, remove the node from the domain and adjust L accordingly. To handle melting at the start of the domain, use a 'ghost' node as the first boundary (**Figure 5**). The ghost node adds an additional 0.5 m at the start of the domain to enforce the BC temperature of 2 °C. The steps to simulate the temperature in the melting glacier are illustrated in **Figure 5**.

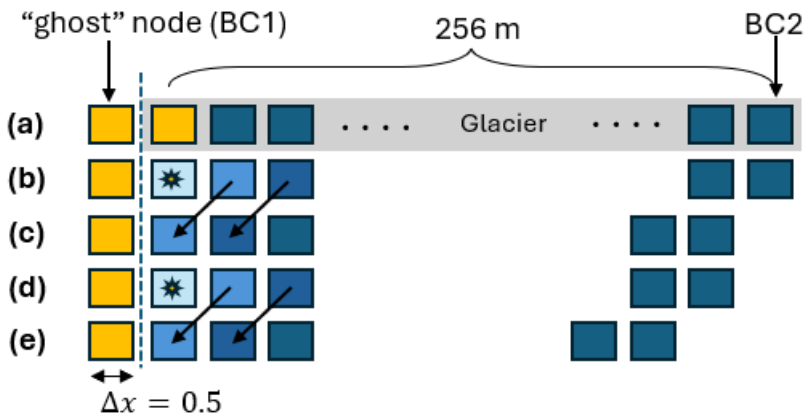

Figure 5. Concept for implementing the melting glacier numerically.

The sequence shown in **Figure 5** is as follows:

(a) This is the IC. Here the ghost node (not the first node of the glacier) enforces the BC.
(b) Eventually the node with the 'star' reaches the melting condition.
(c) Removing the melted node brings the remaining nodes closer to the start, shortening the glacier by Δx. This in turn becomes the new IC to re-start the numerical simulation.

The process then repeats in **(d)** and **(e)**.

- To implement these steps in MATLAB, consider writing a script that calls a modified version of your main function from Q2. What inputs should the modified function have to make your implementation easier?
- When plotting, keep the domain 256 m so that it is easier to compare with the non-melting case. Any parts of the glacier that have melted away can be plotted as 2 °C.

- Because you'll end up with 8 curves to plot (2 each for $t = 10$ d, $t = 100$ d, $t = 1000$ d, and $t = 10000$ d), consider making the 2 curves at a given time the same colour, with one a dashed line and the other an unbroken line. This will simplify visualisation compared to using 8 different colours.

3.2 Report the length of glacier that has melted at $t = 10, 100, 1000$ and 10000 d. What if the environmental temperature rose to 3 °C, how does the amount of melting compare? **[2 mk]**

3.3 Briefly discuss the findings and implications from your simulation. **[2 mk]**

APPENDIX

Error norms are computed as in the lecture:

$$L^p = \left(\sum_{i=1}^{n_x} |f_i - f(x_i)|^p \right)^{\frac{1}{p}} (\Delta x)^{\frac{1}{p}} \tag{9}$$

where f_i is the numerical approximation and $f(x_i)$ is the analytical solution at $x = x_i$, and $p = 1, 2, \infty$. You can use the MATLAB 'norm' function to help you compute error norms.

Convergence rate is then computed as:

$$\mathcal{O}(L^p) = \frac{\log\left(\frac{L_A^p}{L_B^p}\right)}{\log\left(\frac{\Delta x_A}{\Delta x_B}\right)} \tag{10}$$

where A and B refer to two different grid choices.